

Book 2: Chapter 5 – Financial Labels

Machine Learning for Asset Managers (Marcos López de Prado)

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Chapter 5 Overview: The Labeling Problem

Why Standard Prediction Fails

- **Low Signal-to-Noise Ratio:** Raw price returns $r_{t,t+1}$ are dominated by noise and microstructure frictions.
- **Non-Stationarity:** Fixed-horizon price targets yield non-stationary label distributions across varying volatility regimes.
- **Supervised ML Goal:** Transform continuous price series into structured target labels (y_i) representing real investment outcomes.

Chapter 5 Roadmap

- ① **Fixed-Horizon Method:** Academic baseline and its fundamental flaws.
- ② **Triple-Barrier Method:** Path-dependent labeling with profit-taking, stop-loss, and expiration.
- ③ **Trend-Scanning Method:** Dynamic-horizon trend labeling via OLS t -value maximization.
- ④ **Meta-Labeling & Bet Sizing:** Decoupling direction from bet sizing ($m \in [-1, 1]$).

Topic 5.1: The Fixed-Horizon Method & Its Flaws

Traditional Academic Formulation

Sample features X_i at time $t_{i,0}$ and calculate return over fixed horizon h :

$$r_{t_{i,0}, t_{i,0}+h} = \frac{p_{t_{i,0}+h} - p_{t_{i,0}}}{p_{t_{i,0}}}$$
$$y_i = \begin{cases} +1 & \text{if } r_{t_{i,0}, t_{i,0}+h} > \tau \\ -1 & \text{if } r_{t_{i,0}, t_{i,0}+h} < -\tau \\ 0 & \text{otherwise} \end{cases}$$

Three Critical Flaws in Practice

- 1 **Volatility Inconsistency:** Fixed threshold τ produces mostly 0s during quiet regimes, but random ± 1 s during volatile regimes (breaks stationarity).
- 2 **No Intra-Horizon Stops:** Assumes holding blindly until $t_{i,0} + h$, ignoring intermediate stop-loss/profit-taking.
- 3 **Path Independence Failure:** Ignores drawdown and liquidation risk during holding period.

Topic 5.2: The Triple-Barrier Method

Path-Dependent Labeling Architecture

Defines three dynamic exit boundaries based on rolling EWMA volatility σ_t :

- 1 **Upper Horizontal Barrier (π_+)**: Profit-taking limit ($\pi_+ = p_0 \cdot [1 + \theta_1 \sigma_t]$).
- 2 **Lower Horizontal Barrier (π_-)**: Stop-loss limit ($\pi_- = p_0 \cdot [1 - \theta_2 \sigma_t]$).
- 3 **Vertical Barrier (t_1)**: Maximum holding period / expiration ($t_1 = t_0 + H$).

Label Assignment Logic

$$y_i = \begin{cases} +1 & \text{if Upper Barrier touched first (Profit-Taking)} \\ -1 & \text{if Lower Barrier touched first (Stop-Loss)} \\ 0 \text{ or } \text{sgn}(r_{t_0, t_1}) & \text{if Vertical Barrier touched first (Expiration)} \end{cases}$$

Topic 5.3: The Trend-Scanning Method

Motivation: Letting Trends Run

Avoids setting arbitrary fixed horizons h or rigid barrier parameters (θ_1, θ_2) . Identifies prevailing market trends and lets them persist as long as statistically supported.

Linear Regression t -Value Formulation

For entry time t_0 , evaluate OLS regressions over forward lookahead windows $L \in [\underline{L}, \bar{L}]$:

$$p_{t_0+l} = \alpha + \beta \cdot l + \epsilon_l, \quad l = 0, 1, \dots, L$$

Select optimal horizon L^* that maximizes the absolute t -value of slope coefficient $\hat{\beta}$:

$$L^* = \arg \max_{L \in [\underline{L}, \bar{L}]} |t_{\hat{\beta}}(L)|$$

Label $y_i = \text{sgn}(\hat{\beta}_{L^*})$ (or continuous $t_{\hat{\beta}}(L^*)$) reflects trend sign and statistical strength.

Comparing Financial Labeling Methodologies

Feature	Fixed-Horizon	Triple-Barrier	Trend-Scanning
Path-Dependent?	No	Yes	Yes
Volatility Adaptive?	No	Yes (σ_t -scaled)	Indirectly (t -stat)
Parameters Needed	h, τ	θ_1, θ_2, H	$[\underline{L}, \bar{L}]$
Handles Stop-Loss?	No	Yes (Explicit)	No (Exit on trend break)
ML Target Type	Discrete $\{-1, 0, 1\}$	Discrete $\{-1, 0, 1\}$	Discrete or Continuous t -stat
Primary Use Case	Academic Benchmark	Risk-Managed Trading	Trend-Following Strategies

Topic 5.4: Meta-Labeling Architecture

Decoupling Trade Direction (Side) from Position Size (Size)

Standard classifiers attempt to learn direction and size simultaneously. Meta-labeling splits the problem into two distinct stages:

1. Primary Model

- Generates trade **Side** $s_i \in \{-1, +1\}$ (Buy/Sell).
- Can be a white-box model: MACD, fundamental value, or PM discretion.
- Optimized for **High Recall** (capturing opportunities).

2. Secondary ML Model

- Applies Triple-Barrier **only in primary direction**.
- Target: $y_i^{\text{meta}} = 1$ if profit-taking hit (True Positive); 0 if loss/expiration.
- Predicts probability $p = P[y^{\text{meta}} = 1]$ to size bets.

The Four Superpowers of Meta-Labeling

- ➊ **Precision Optimization (Precision Lift):** Sacrifices a minor amount of recall to eliminate false positives, dramatically raising trading win rate (precision) and Sharpe ratio.
- ➋ **Quantamental Integration:** Allows discretionary/fundamental PMs (white box) to overlay an ML risk filter (black box) to filter bad calls without giving up economic rationale.
- ➌ **Overfitting Control:** ML model does not decide market direction (which is extremely noisy), but only whether to execute and how much to bet.
- ➍ **Asymmetric Long/Short Sizing:** Enables separate ML sizing models for long vs. short regimes, reflecting structural market asymmetries.

Topic 5.5: Bet Sizing by Expected Sharpe Ratio

Converting Probability p into Position Size m

Let $p = P[y_i^{\text{meta}} = 1]$ be predicted probability of success. Assuming profit π and loss $-\pi$, the expected Sharpe ratio for a trade maps to continuous position size $m \in [-1, 1]$:

$$m = 2Z \left[\frac{p - 1/2}{\sqrt{p(1-p)}} \right] - 1$$

where $Z[\cdot]$ is the CDF of the standard normal distribution.

Key Properties

- When $p = 0.5$ (no edge), $m = 2Z[0] - 1 = 0$ (zero bet size / no trade).
- As $p \rightarrow 1.0$ (high confidence), $m \rightarrow +1.0$ (maximum allocation).

Topic 5.5: Ensemble Bet Sizing

Averaging n Independent Classifiers

Consider n meta-labeling classifiers making binary predictions. Under $H_0 : p = 1/2$ (no predictive power), the test statistic t follows Student's t -distribution with $n - 1$ degrees of freedom:

$$t = \frac{\hat{p} - 1/2}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$$

Statistically Grounded Bet Size

$$m = 2F_{t,n-1}(t) - 1 \in [-1, 1]$$

where $F_{t,n-1}(\cdot)$ is the CDF of Student's t -distribution with $n - 1$ degrees of freedom.

Summary: Key Takeaways on Financial Labels

- ❶ **Abandon Fixed Horizons:** Standard h -bar labeling introduces non-stationarity and ignores intermediate path-dependent liquidation risk.
- ❷ **Use Path-Dependent Barriers:** The Triple-Barrier Method mirrors real-world trading rules by incorporating profit-taking, stop-loss, and vertical expiration.
- ❸ **Capture Persistent Trends:** The Trend-Scanning Method uses OLS t -value maximization for dynamic-horizon trend labeling.
- ❹ **Leverage Meta-Labeling:** Decouple direction from bet sizing. Use primary models for side and secondary ML models for precision filtering and bet sizing ($m \in [-1, 1]$).

Thank You!

Questions & Discussion

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